Course One
Foundations of Data Science

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- ☐ Complete the questions in the Course 1 PACE strategy document
- ☐ Answer the questions in the Jupyter notebook project file
- ☐ Complete coding prep work on project's Jupyter notebook
- ☐ Summarize the column Dtypes
- ☐ Communicate important findings in the form of an executive summary

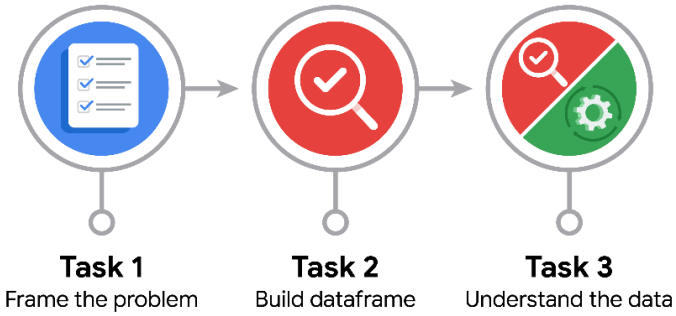
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

The "Context" Filter: Read through the emails from Rosie and Orion twice. Once for **technical requirements** (data types, null counts) and once for **business goals** (reducing moderation backlog).

Glossary Building: Define "Claim" vs. "Opinion" immediately. These are your target classes for the future model.

- What follow-along and self-review codebooks will help you perform this work?

Course 2 Python & Pandas Notebooks: Specifically those focusing on the `pandas` library, data cleaning, and inspecting dataframes (`.info()`, `.describe()`, `.shape`).

Data Exploration Exemplars: Looking at previous project solutions to see how professionals structure their "Executive Summaries.

- What are some additional activities a resourceful learner would perform before starting to code?

Variable Mapping: Create a list of which columns are "signals" (like view count) and which are "noise" (like a random row index).

Documentation Review: Check the Pandas documentation for `df.agg()` or `df.groupby()` to prepare for Orion's request to create "meaningful variables."

PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Intuition: Likely yes, *if* the data includes user engagement (likes/shares) and account status (verified). However, we must ensure the "Claim" labels are balanced. If 99% of the data is "Opinions," the model won't learn what a "Claim" looks like.

- How would you build summary dataframe statistics and assess the min and max range of the data?

Use the `df.describe()` method. It automatically calculates the `min`, `max`, `mean`, and `quartiles` for all numerical columns. For categorical columns, use `df['column'].value_counts()`.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

Unusual Averages: Look for "Extreme Means." If the average `video_like_count` is 500, but the maximum is 5,000,000, the data is heavily **skewed**. **Interval Data:** This refers to data where the distance between points is equal (like counts or timestamps). You'll want to check if the time ranges for video uploads are consistent or if there are gaps in the data.

PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

Verification Bias: I would recommend checking if "verified" accounts are more likely to post opinions than claims.

Source of Reports: Investigating *who* reports the videos. Are certain users reporting every opinion they disagree with as a "false claim"?

- What data initially presents as containing anomalies?

Zero-Engagement Videos: Videos with high view counts but 0 likes or 0 shares are often bot traffic or technical glitches.

Outliers: Any video with millions of views but only a few seconds of watch time.

- What additional types of data could strengthen this dataset?

Transcription/NLP Data: Having the actual text of what was said in the video (via speech-to-text) would be the most powerful predictor for a claim-vs-opinion model.

User History: Has the user who posted the video been banned or flagged before? (`author_ban_status` is helpful here).